

Data Science Capstone- SpaceX

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OUTLINE

- INTRODUCTION
- SUMMARY
- METHODOLOGY
- RESULTS
- CONCLUSION
- APPENDIX



INTRODUCTION

Project Background:

SpaceX advertises Falcon 9 rocket launches on its website, with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.

Problem:

Predicting if the Falcon 9 first stage will land successfully.

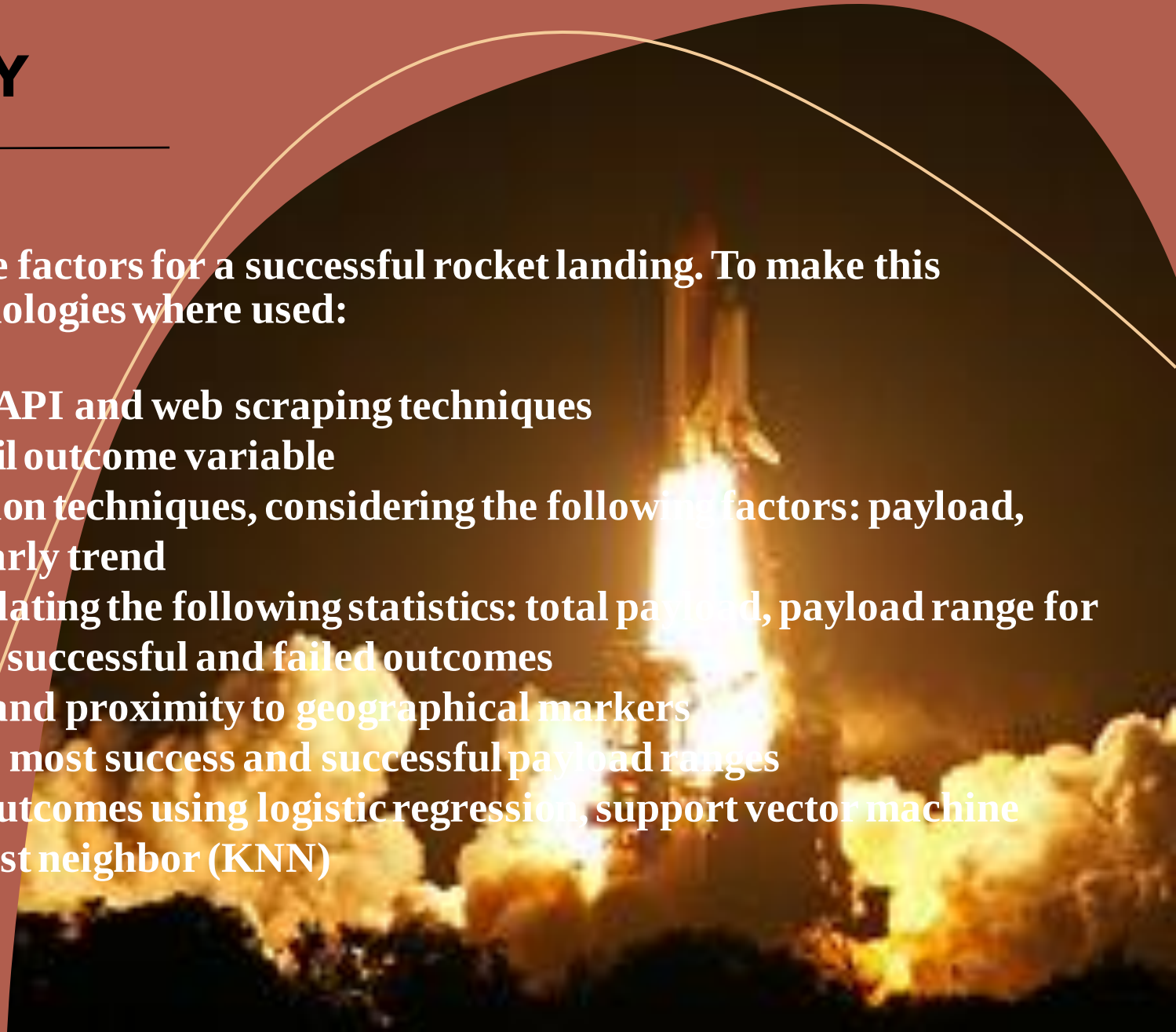


EXECUTIVE SUMMARY

- **Methodologies Summary**

The research attempts to identify the factors for a successful rocket landing. To make this determination, the following methodologies were used:

- Collect data using SpaceX REST API and web scraping techniques
- Wrangle data to create success/fail outcome variable
- Explore data with data visualization techniques, considering the following factors: payload, launch site, flight number and yearly trend
- Analyze the data with SQL, calculating the following statistics: total payload, payload range for successful launches, and total # of successful and failed outcomes
- Explore launch site success rates and proximity to geographical markers
- Visualize the launch sites with the most success and successful payload ranges
- Build Models to predict landing outcomes using logistic regression, support vector machine (SVM), decision tree and K-nearest neighbor (KNN)



EXECUTIVE SUMMARY

- **Results Summary**

Exploratory Data Analysis:

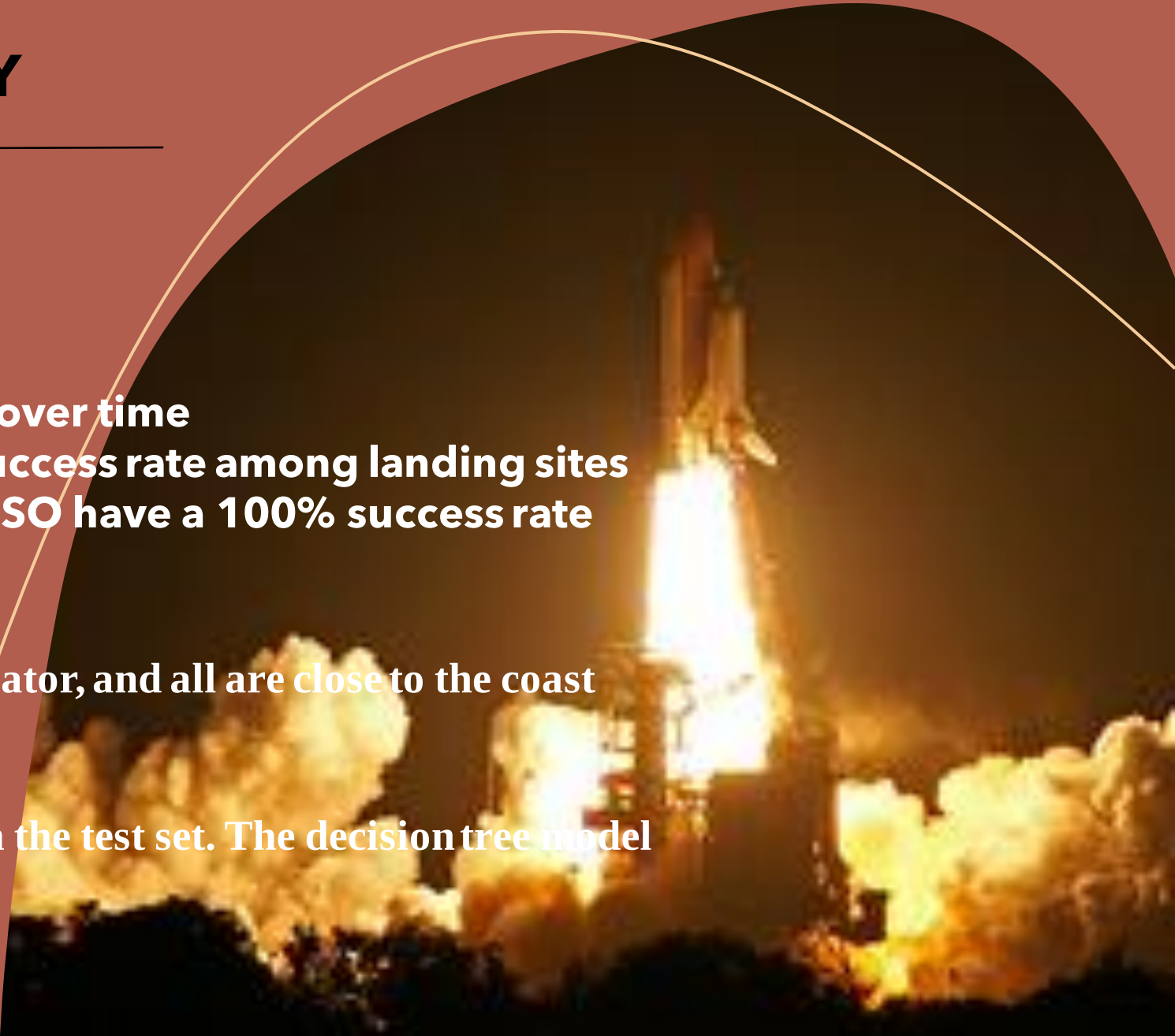
- **Launch success has improved over time**
- **KSC LC-39A has the highest success rate among landing sites**
- **Orbits ES-L1, GEO, HEO, and SSO have a 100% success rate**

Visualization / Analytics:

- **Most launch sites are near the equator, and all are close to the coast**

Predictive Analytics

- **All models performed similarly on the test set. The decision tree model**



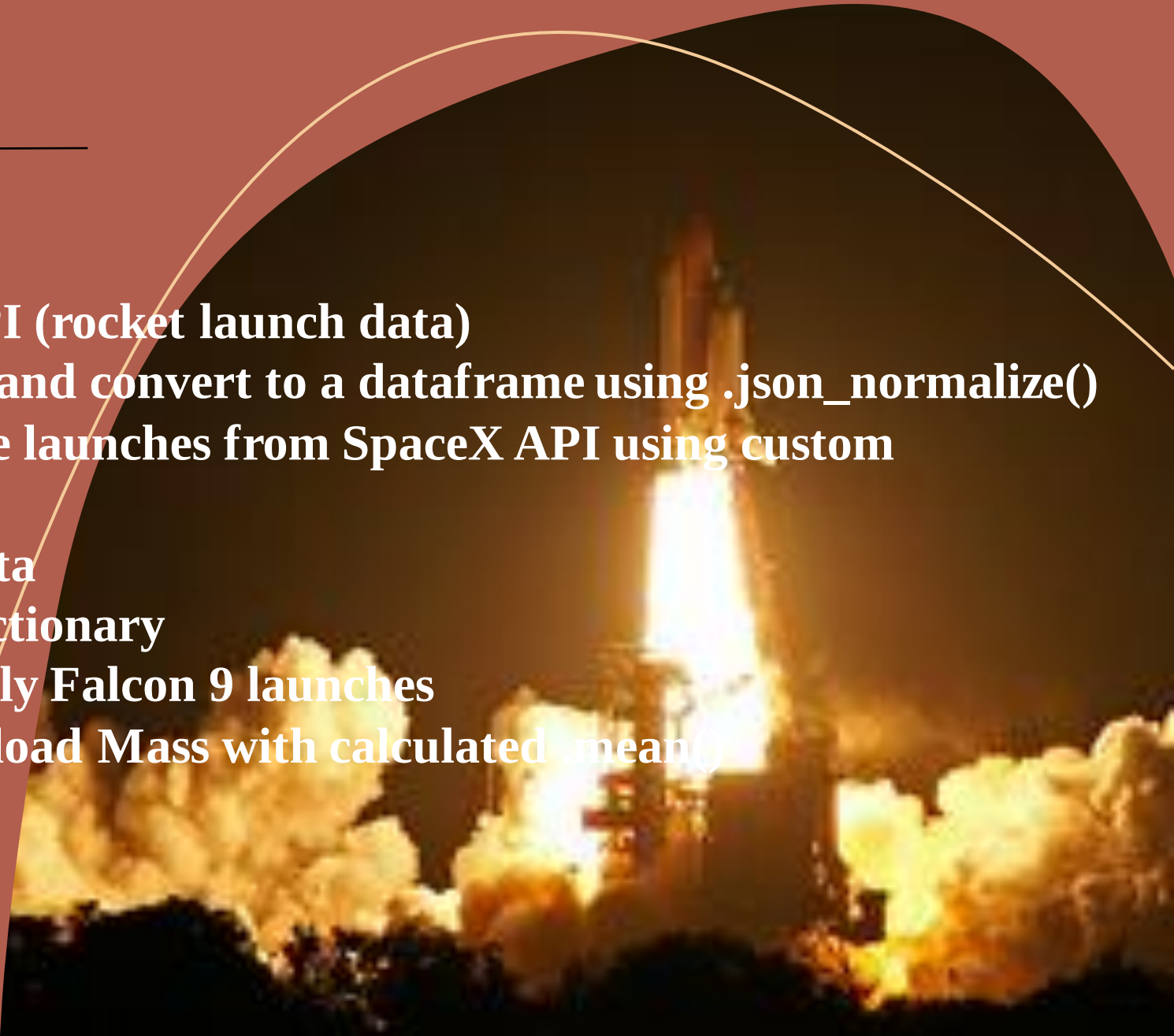
METHODOLOGY

- Data Collection
- Data Wrangling
- EDA Data visualization.
- EDA Sql data
- Machine Learning Prediction
- Interactive Visual Analytics with Folium
- Build a Dashboard Using Plotly Dash



METHODOLOGY

- Data Collection
 - Request data from SpaceX API (rocket launch data)
 - Decode response using `.json()` and convert to a dataframe using `.json_normalize()`
 - Request information about the launches from SpaceX API using custom functions
 - Create dictionary from the data
 - Create dataframe from the dictionary
 - Filter dataframe to contain only Falcon 9 launches
 - Replace missing values of Payload Mass with calculated `.mean()`
 - Export data to csv file



METHODOLOGY

- **Data Collection**

SpaceX API:

- The SpaceX API has data available publically.
- Once a GET request has been made to the SpaceX API and the response received, the data can be placed into a Pandas DataFrame for further analysis.
- **Link:** <https://github.com/REDHWAN0/IBM-Data-Science-Capstone-SpaceX/Data-collection-api.ipynb>

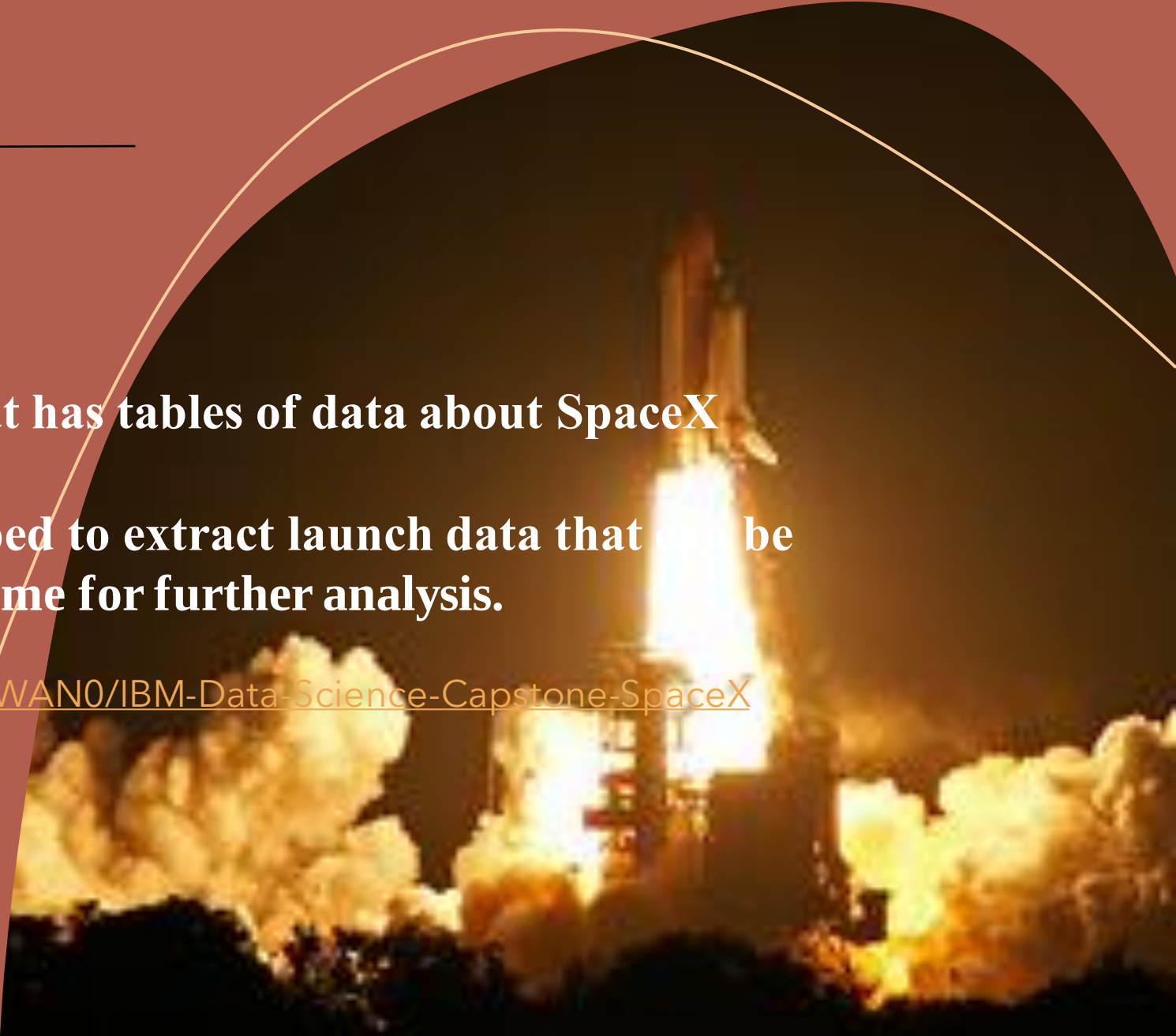


METHODOLOGY

- **Data Collection**

Scraping:

- **Wikipedia** has a page that has tables of data about SpaceX launches.
- These tables can be scraped to extract launch data that can be put into a Pandas DataFrame for further analysis.
- **Link:** [https://github.com/REDH/WAN0/IBM-Data-Science-Capstone-SpaceX/Web Scraping.ipynb](https://github.com/REDH/WAN0/IBM-Data-Science-Capstone-SpaceX/Web%20Scraping.ipynb)



METHODOLOGY

- **Data Wrangling**
 - Convert outcomes into 1 for a successful landing and 0 for an unsuccessful landing
 - The .csv file contains the data that needed to be cleaned.
 - The launch sites, orbit types and mission outcomes were cleaned up.
 - The new classification was added to the DataFrame for further analysis
 - Filtering the data
 - Dealing with missing values
 - Using One Hot Encoding to prepare the data to a binary classification
- **Link:** [https://github.com/REDHWAN0/IBM-Data-Science-Capstone-SpaceX/Data wrangling.ipynb](https://github.com/REDHWAN0/IBM-Data-Science-Capstone-SpaceX/Data%20wrangling.ipynb)



METHODOLOGY

- **EDA with Data Visualization**

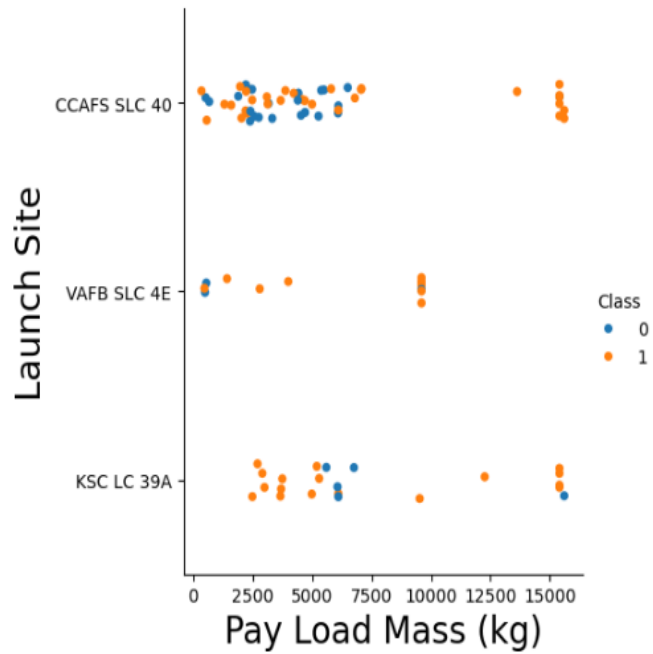
The following charts were created to look at Launch Site trends :

- **Scatterplot to see mission outcome relationship split by Launch Site and Flight Number.**
- **Scatterplot to see mission outcome relationship split by Launch Site and Payload.**

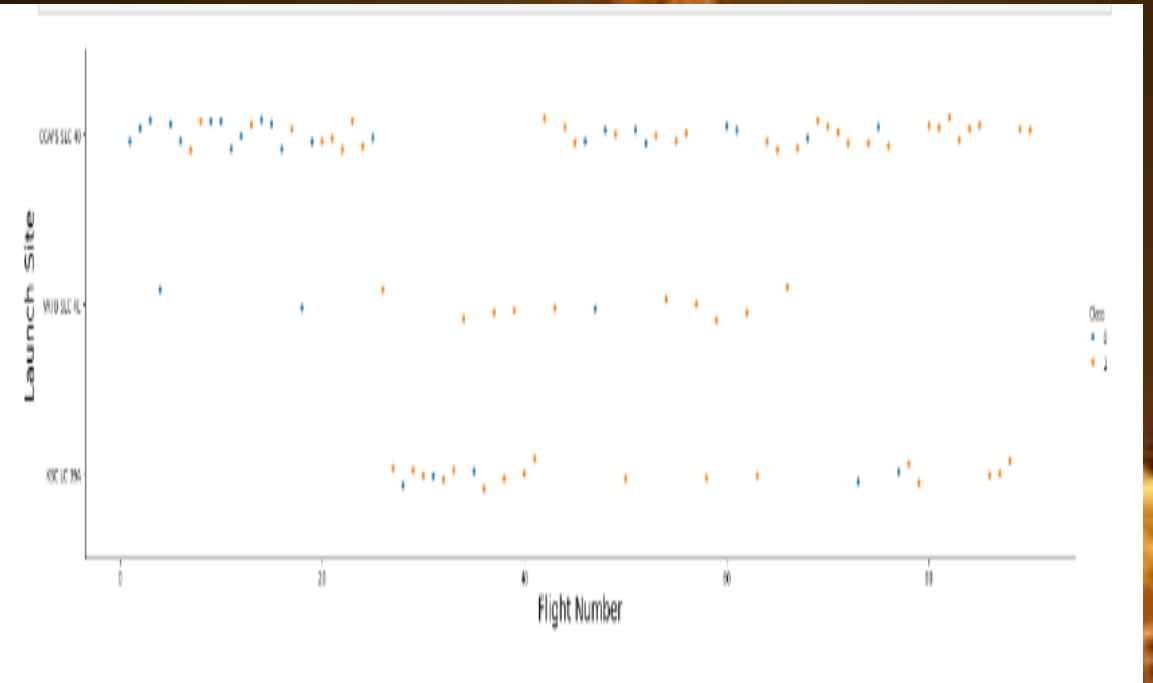


METHODOLOGY

- EDA with Data Visualization



Activate Windows
Go to Settings to activate Windows

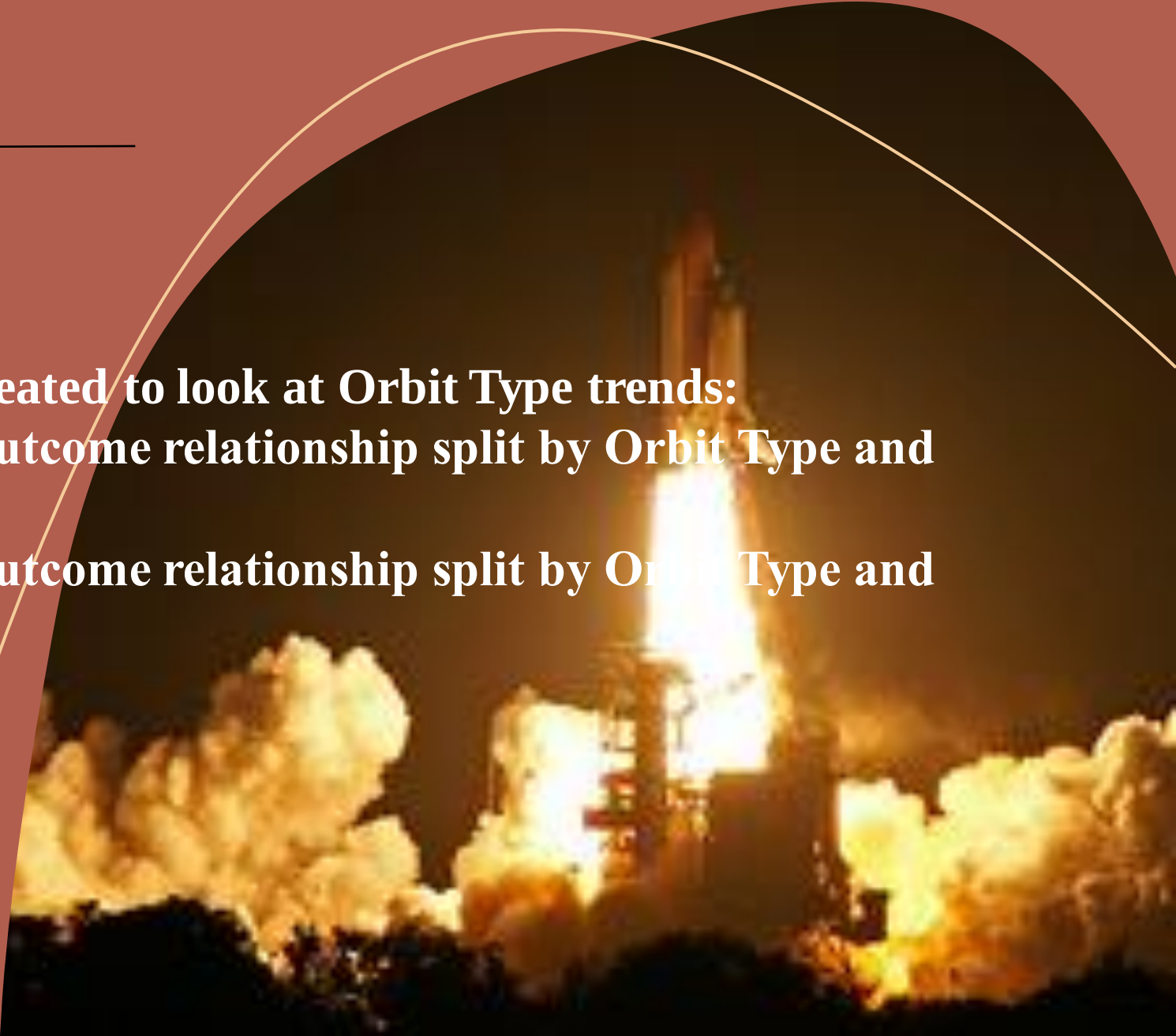


METHODOLOGY

- EDA with Data Visualization

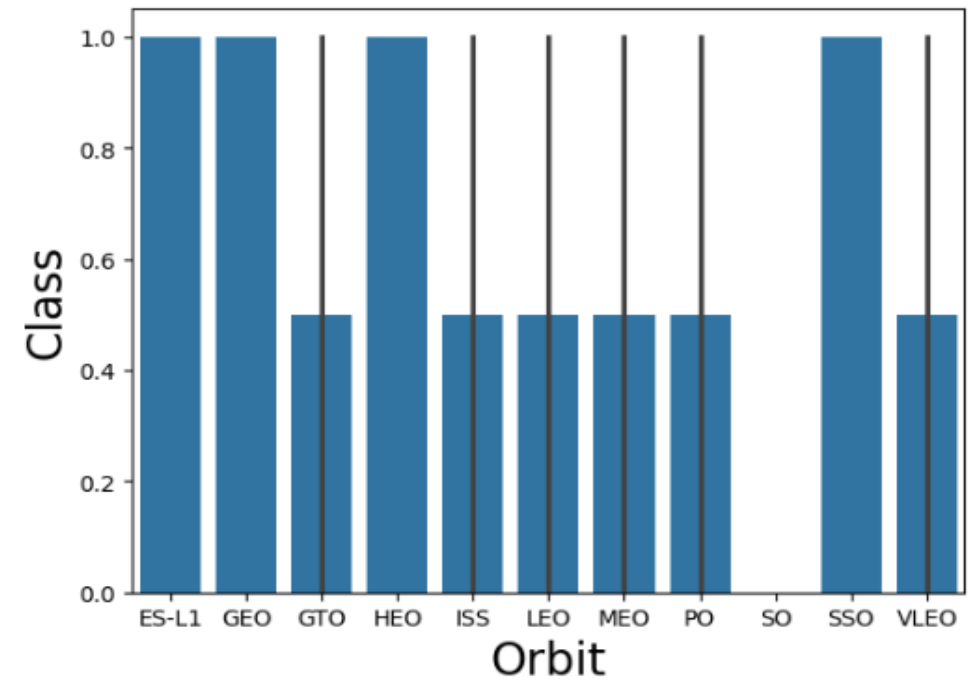
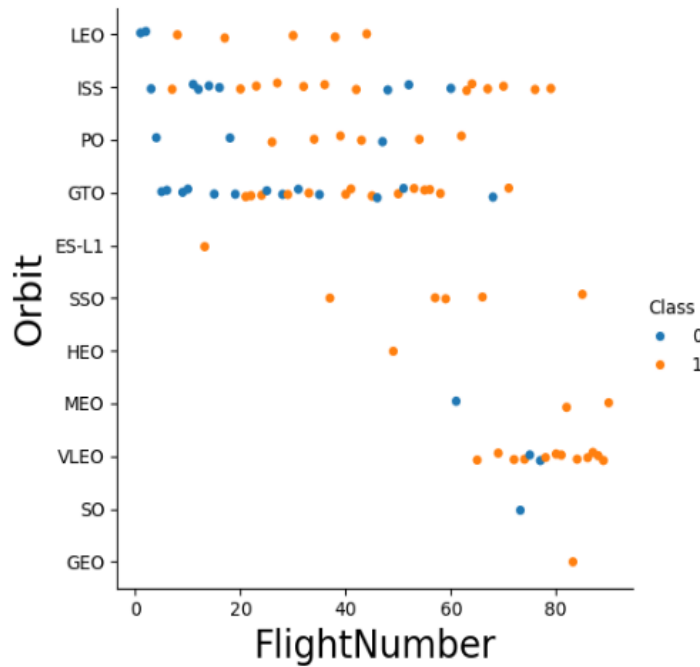
The following charts were created to look at Orbit Type trends:

- Scatterplot to see mission outcome relationship split by Orbit Type and Flight Number.
- Scatterplot to see mission outcome relationship split by Orbit Type and Payload.



METHODOLOGY

- EDA with Data Visualization

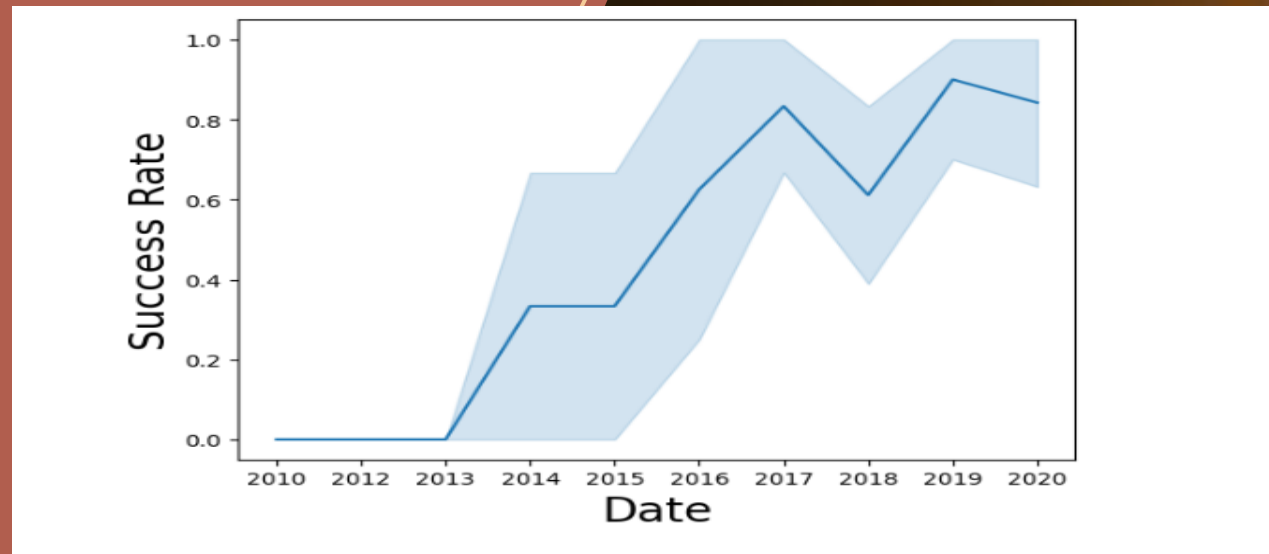


METHODOLOGY

- EDA with Data Visualization

The following chart was created to look at trends based on time:

- Line plot to see mission outcome trend by year.



- **GitHub URL (EDA with Data Visualization):** : [https://github.com/REDHIWAN0/IBM-Data-Science-Capstone-SpaceX/EDA-Data visualization.ipynb](https://github.com/REDHIWAN0/IBM-Data-Science-Capstone-SpaceX/EDA-Data%20visualization.ipynb)

METHODOLOGY

- EDA Sql data

Queries were written to extract information about:

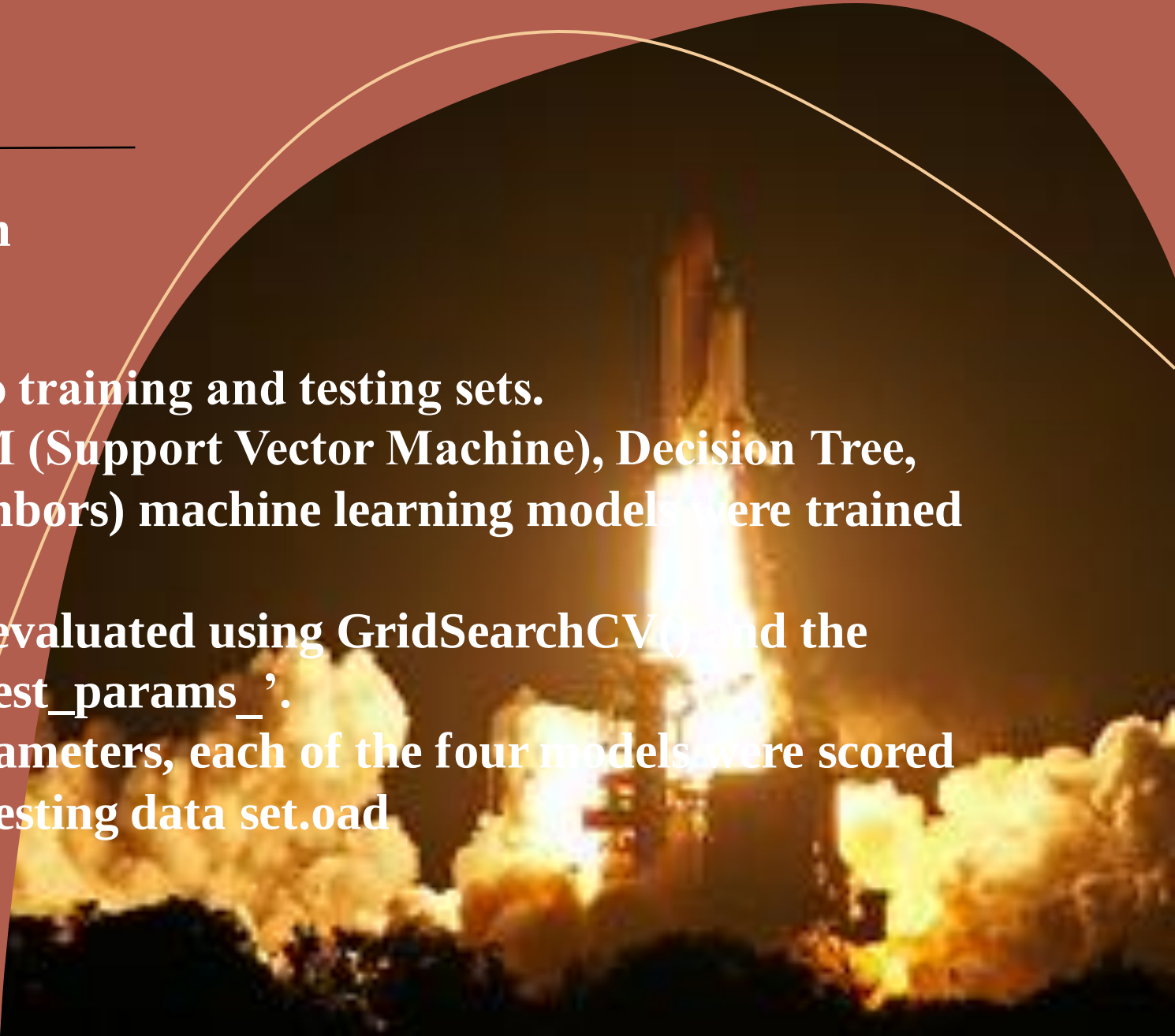
- Launch sites
- Payload masses
- Dates
- Booster types
- Mission outcomes

- **GitHub URL (EDA Sql data)** : <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX%20EDA-Sql%20data.ipynb>



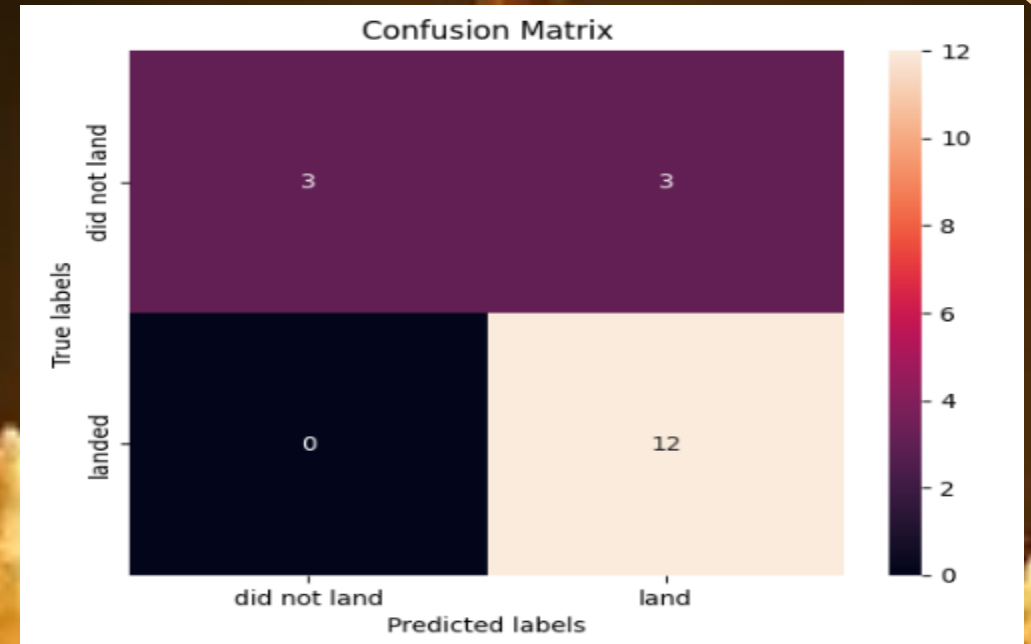
METHODOLOGY

- **Machine Learning Prediction**
 - The dataset was split into training and testing sets.
 - Logistic Regression, SVM (Support Vector Machine), Decision Tree, and KNN (k-Nearest Neighbors) machine learning models were trained on the training data set.
 - Hyper-parameters were evaluated using GridSearchCV() and the best was selected using 'best_params_'.
 - Using the best hyper-parameters, each of the four models were scored on accuracy by using the testing data set.



METHODOLOGY

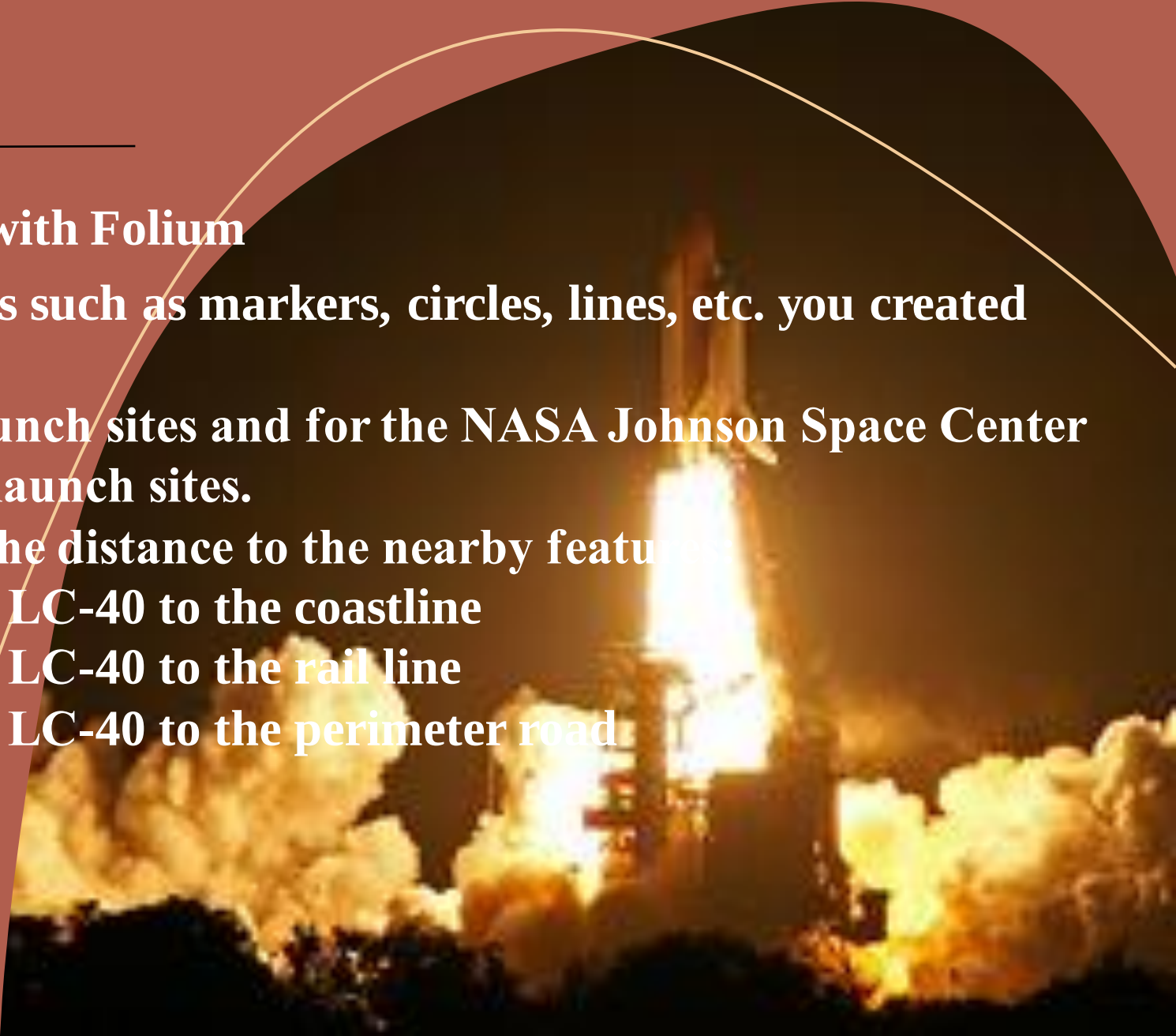
- **Machine Learning Prediction:**
 - Shown here is the confusion matrix for the Logistic Regression model.
 - Prediction Breakdown:
 - 12 True Positives and 3 True Negatives
 - 3 False Positives and 0 False Negatives



- **GitHub URL (Machine Learning Prediction) :** [https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX%20Machine Learning Prediction.ipynb](https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX%20Machine%20Learning%20Prediction.ipynb)

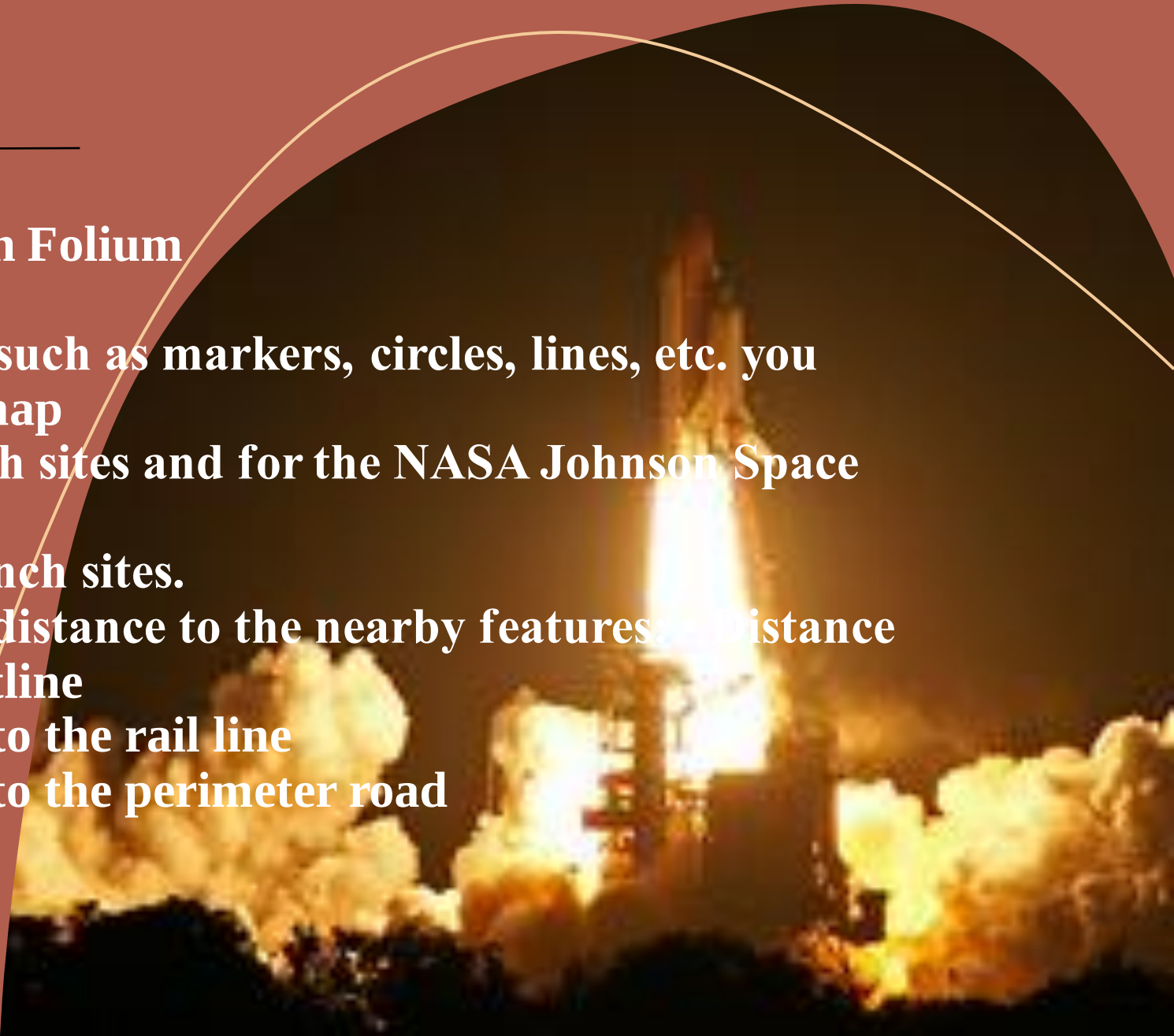
METHODOLOGY

- **Interactive Visual Analytics with Folium**
Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
 - Markers were added for launch sites and for the NASA Johnson Space Center
 - Circles were added for the launch sites.
 - Lines were added to show the distance to the nearby features
 - Distance from CCAFS LC-40 to the coastline
 - Distance from CCAFS LC-40 to the rail line
 - Distance from CCAFS LC-40 to the perimeter road

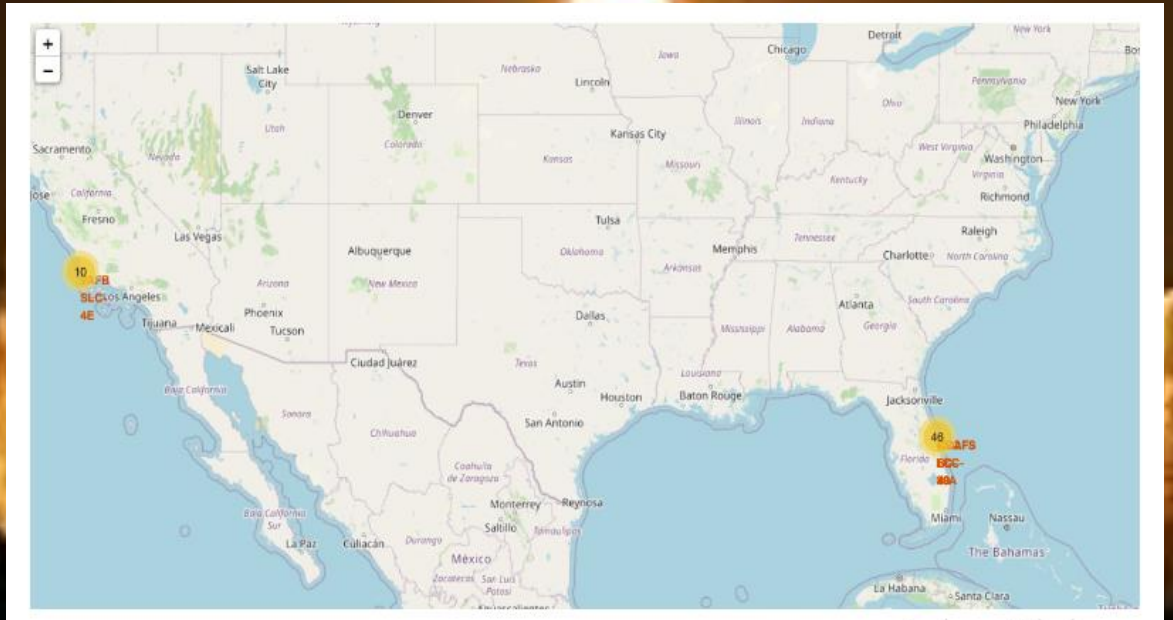


METHODOLOGY

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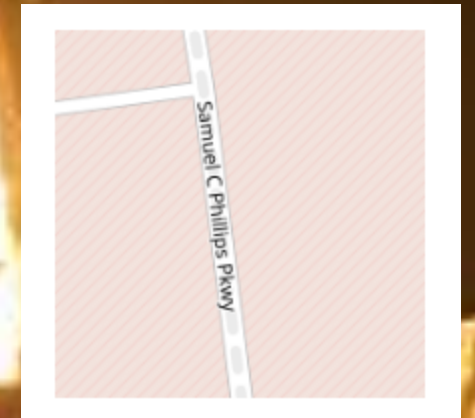
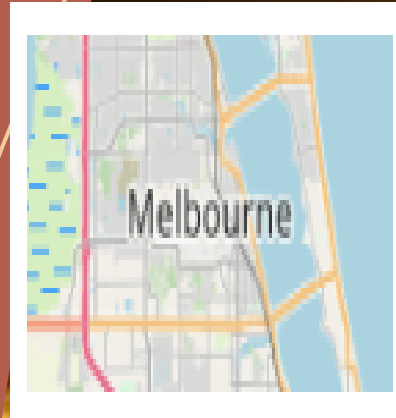
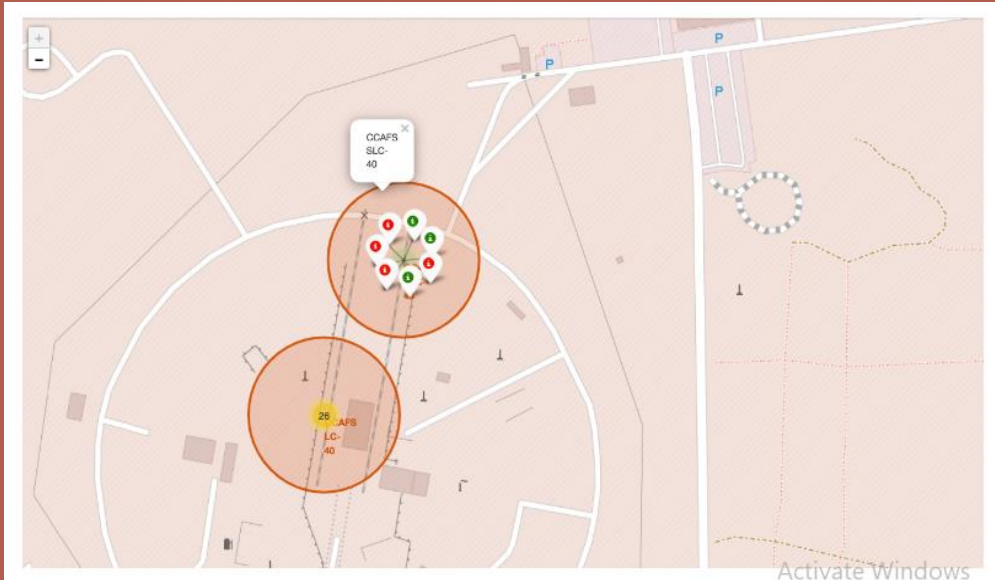


- **Interactive Visual Analytics with Folium**



METHODOLOGY

- Interactive Visual Analytics with Folium



- **GitHub URL (Interactive Visual Analytics with Folium)** : <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX%20Interactive%20Visual%20Analytics%20with%20Folium.ipynb>

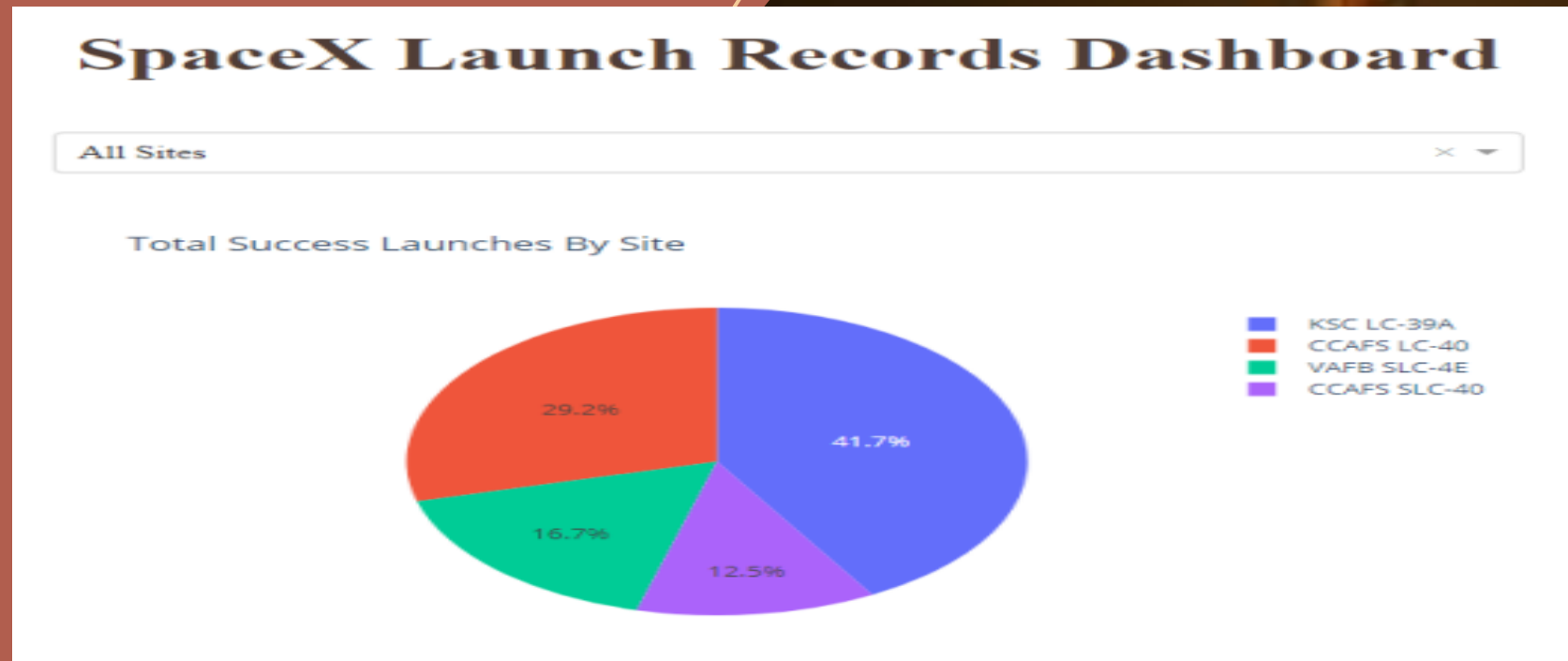
METHODOLOGY

- **Build a Dashboard Using Plotly Dash**
 - The input dropdown is used to select one or all launch sites for the pie chart and scatterplot.
 - The pie chart displays one of two things:
 - For All Sites – the distribution of successful Falcon 9 first stage landings between the sites
 - For One Site – the distribution of successful and failed Falcon 9 first stage landings for that site
 - The input slider is used to filter the payload masses for the scatterplot.
 - The scatterplot displays the distribution of Falcon 9 first stage landings split by payload mass, mission outcome and by booster version category.



METHODOLOGY

- Build a Dashboard Using Plotly Dash



- **GitHub URL (Build a Dashboard Using Plotly Dash) :** https://github.com/RADHWANO/IBM-Data-Science-Capstone-SpaceX%20/plotly_dash.py

RESULTS

Exploratory Data Analysis:

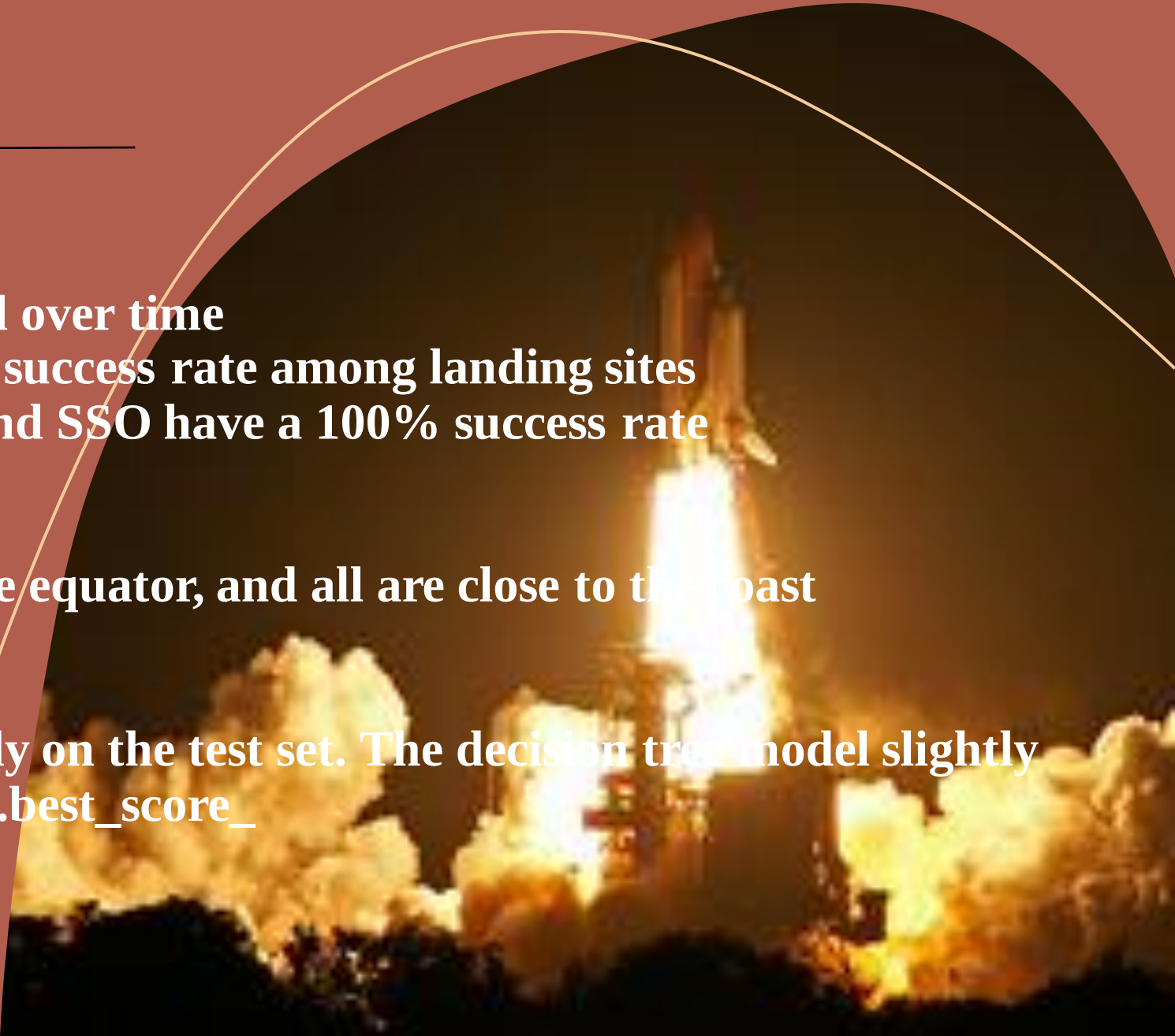
- Launch success has improved over time
- KSC LC-39A has the highest success rate among landing sites
- Orbits ES-L1, GEO, HEO, and SSO have a 100% success rate

Visualization / Analytics:

- Most launch sites are near the equator, and all are close to the coast

Predictive Analytics

- All models performed similarly on the test set. The decision tree model slightly outperformed when looking at `.best_score_`



CONCLUSION

- SpaceX does not have a perfect track record of Falcon 9 first stage landing outcomes.
- SpaceX's Falcon 9 first stage landing outcomes have been trending towards greater success as more launches are made.
- The machine learning models can be used to predict future SpaceX Falcon 9 first stage landing outcomes



APPENDIX

Initial Data Sets

- Wikipedia

(Webpage): https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922



APPENDIX

- **Data Sets (.csv files):**
- **GitHub URL (CSV 1):**
https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /dataset_part_1.csv
- **GitHub URL (CSV 2):**
https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /dataset_part_2.csv
- **GitHub URL (CSV 3):**
https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /dataset_part_3.csv
- **GitHub URL (SpaceX):**
<https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /SpaceX.csv>
- **GitHub URL (Launch Geo):**
https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /plotly_dash.py
- **GitHub URL (Plotly Dash):**
https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /plotly_dash.py/



APPENDIX

Jupyter Notebooks and Dashboard Python File

- **GitHub URL (Data Collection):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /Data-collection-api.ipynb>
- **GitHub URL (Web Scraping):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /Web Scraping.ipynb>
- **GitHub URL (Data Wrangling):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /Data wrangling.ipynb>
- **GitHub URL (EDA Sql data):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /EDA-Sql data.ipynb>
- **GitHub URL (EDA with Data Visualization):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /EDA-Data visualization.ipynb>
- **GitHub URL (Folium Maps):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /Interactive Visual Analytics with Folium.ipynb>
- **GitHub URL (Plotly Dash):** https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /plotly_dash.py
- **GitHub URL (Machine Learning):** <https://github.com/RADHWAN0/IBM-Data-Science-Capstone-SpaceX /Machine Learning.ipynb>